HANANEEL EPHRAIM MANDAM

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EDUCATION

B.E. Mechanical Engineering | BITS Pilani, Hyderabad Campus *Expected 2028*

Coursework: Design of Machine Elements, Advanced Mechanics of Solids, Mechanisms & Machines, Vibrations & Control, Fluid Mechanics, Applied Thermodynamics, Heat Transfer, Principles of Aerodynamics, Engines, Motors & Mobility, Materials Science, AI for Robotics, Foundations of Data Science, Differential Equations

EXPERIENCE

Research Intern, Physiological Signal Processing | CSIR-CEERI, Chennai *May 2026 – Jul 2026*

- Raised R-peak detection accuracy to **98.8%** vs. **81.7%** for NeuroKit2 on the MIT-BIH NSTDB noise-stress benchmark (24 dB and 18 dB SNR) by improving Pan-Tompkins++, cutting missed and false heartbeats in noisy real-world signals.
- Delivered an end-to-end real-time ECG/HRV pipeline for a defense project, estimating KSS drowsiness levels from live commercial health-band data over rolling windows.
- Enabled multi-subject live sessions through a pub/sub ECGBus architecture over BLE, with hardware-level signal-quality gating on Polar H10 straps and automatic logging of vehicle and software crash events.

Powertrain Engineer & Electrical Lead | Electric Baja SAE, BITS Pilani Hyderabad *Apr 2025 – Apr 2026*

- Caught a fillet below a safety factor of **1.0** in ANSYS FEA at **90 Nm** peak torque, a part that would have failed under full load, and iterated the geometry before fabrication; redesigned the drivetrain into a manufacturable **4:1** spur stage plus chain-sprocket final drive after the first gearbox proved unbuildable with team tooling, sizing gears from free-body diagrams with Lewis and AGMA hand calcs.
- Delivered the vehicle safety interlock from scratch to SAE E-Baja rules (microcontroller code, relay logic, HV/LV interface); validated shutdown on every failure case forced on the bench and documented the full wiring schematic.
- Coordinated a **7-person** team through fabrication and vendor timelines; profiled and welded chassis members for the electric drivetrain conversion.

PROJECTS

Uncertainty-Aware UAV Emergency Landing Zone Selection | GitHub *Jun 2026 – Present*

- Achieved **zero** false-safe landing selections with every uncertainty-aware variant on UAVid and Aeroscapes, where temporal consistency alone still marked unsafe zones as safe on both datasets; it never picked unsafe ground as a landing site.
- Built a three-channel risk fusion pipeline: SegFormer-B2 segmentation fine-tuned on UAVid, epistemic uncertainty via MC Dropout, and temporal label-disagreement tracking.

Ground-Effect Wing CFD | Principles of Aerodynamics Coursework *In Progress*

- Matched thin-airfoil theory within **0.3°** on zero-lift angle, with a linear-range lift slope of **0.101/deg** and peak L/D of **62**, using a 2D RANS (k-omega SST) workflow in ANSYS Fluent on an inverted NACA 6412 swept from -12° to +4°.

Self-Landing EDF Hopper: TVC Gimbal & Roll Vanes | Personal Project *In Progress*

- Found and fixed **2** tab collisions before printing through an **11**-configuration interference study, leaving **~1.5 mm** clearance at 10°, so the mechanism no longer jams mid-tilt; resolved a pitch-yaw bind with an hourglass hole.
- Designed a two-axis thrust vector control gimbal in SolidWorks for a 70 mm EDF (**±10°** on both axes via a ~3:1 servo pushrod linkage) plus two direct-drive NACA 0012 roll vanes sized to the jet core.
- Bench-tested servo actuation on an ESP32; next: IMU-driven gimbal demo, then load-cell thrust and roll-authority tests.

Image Stitching for Vision-Based Dimensional Inspection | ME F376 Design Project *Aug 2026 – Present*

- Passed **83/84** cases (< 5 px error) on a ground-truth validation suite I built; it caught a **320 px** silent misalignment on repeating patterns and a regression from my first fix, so the tool refuses rather than report a wrong dimension.
- Raised pixel agreement on a welded plate from **0.58 to 0.94** with background rejection, after diagnosing a stitching artefact as parallax; pipeline built from first principles in Python/OpenCV (ORB, RANSAC/MAGSAC++).

SKILLS

Programming & Tools: Python (PyTorch, NumPy, OpenCV, HuggingFace), MATLAB, Simulink, C, Java, Git, Linux, Excel

CAD, FEA & CFD: SolidWorks, ANSYS (Static Structural, Modal, Fluent), Finite Element Analysis (FEA), Mechanical Design

Structural Analysis: Free-Body Diagrams, Structural Load Analysis, Shear & Moment Diagrams, Lewis/AGMA Gear Rating

ML & Vision: Semantic Segmentation, Uncertainty Quantification, Transfer Learning, Image Registration

Fabrication: MIG/TIG Welding, Lathe, Sheet Metal, FDM 3D Printing, Soldering